

This is a test of making searchable PDF files when using Type 3 bitmap fonts with the plain T_EX, dvips, ps2pdf toolchain.

I've mocked up what I think should happen in `addencodings.pl`. To use it, create a dvi file, process it with `dvips -V1`, stream the PostScript result through `addencodings.pl`, and convert to PDF with `ps2pdf`. The script `fakedvips.pl` does all of this.

Here is some sample text from `testfont`.

On November 14, 1885, Senator & Mrs. Leland Stanford called together at their San Francisco mansion the 24 prominent men who had been chosen as the first trustees of The Leland Stanford Junior University. They handed to the board the Founding Grant of the University, which they had executed three days before. This document—with various amendments, legislative acts, and court decrees—remains as the University's charter. In bold, sweeping language it stipulates that the objectives of the University are “to qualify students for personal success and direct usefulness in life; and to promote the publick welfare by exercising an influence in behalf of humanity and civilization, teaching the blessings of liberty regulated by law, and inculcating love and reverence for the great principles of government as derived from the inalienable rights of man to life, liberty, and the pursuit of happiness.” ¿But aren't Kafka's Schloß and Æsop's Œuvres often naïve vis-à-vis the dæmonic phœnix's official rôle in fluffy soufflés? (¡THE DAZED BROWN FOX QUICKLY GAVE 12345–67890 JUMPS!)

It's unlikely much math will copy/paste legibly, but we can try some simple things and see what happens.

$$a^2 + b^2 = c^2$$

$$F_n = \left\lfloor \frac{\phi^n}{\sqrt{5}} \right\rfloor$$

Font size and Kern vs space tests. Notice the steadily increasing distance between the adjacent ‘b’ and ‘o’; at some point PDF viewers will decide the horizontal space is large enough to treat as a word separator and not a kern. This should probably be somewhere around half the size of a true word space. Also, check that the font size when pasting-with-formatting is roughly correct.

5 point font (cmr5): 0.1 bp: bo. 0.125 bp: bo. 0.15 bp: bo. 0.2 bp: bo. 0.25 bp: bo. 0.3 bp: bo. 0.4 bp: bo. 0.5 bp: bo. 0.6 bp: bo. 0.8 bp: bo. 1 bp: bo. 1.25 bp: bo. 1.5 bp: bo. 2 bp: bo. 2.5 bp: bo. 3 bp: bo. 4 bp: bo. 5 bp: bo. 6 bp: bo. 8 bp: bo. 10 bp: bo. 12.55 bp: bo. 15 bp: bo. 20 bp: bo. 25 bp: bo. 30 bp: bo.

17 point font (cmr10): 0.1 bp: bo. 0.125 bp: bo. 0.15 bp: bo. 0.2 bp: bo. 0.25 bp: bo. 0.3 bp: bo. 0.4 bp: bo. 0.5 bp: bo. 0.6 bp: bo. 0.8 bp: bo. 1 bp: bo. 1.25 bp: bo. 1.5 bp: bo. 2 bp: bo. 2.5 bp: bo. 3 bp: bo. 4 bp: bo. 5 bp: bo. 6 bp: bo. 8 bp: bo. 10 bp: bo. 12.55 bp: bo. 15 bp: bo. 20 bp: bo. 25 bp: bo. 30 bp: bo.